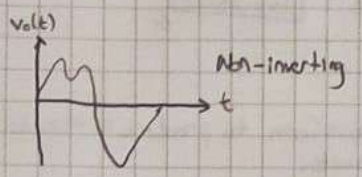
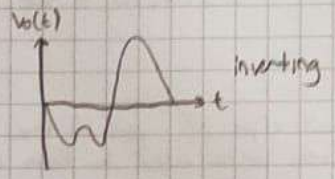
Recep Mepenek Circuit And Electronics HW5

① When the amplifier gain A_v, is negative, the output is an inverted version of the input. This type of amplifier is called inverting amplifier. On the other hand if amplifier gain is positive number we have non-inverting amplifier.





②

$V_{o1} = 100mV$ (closed)
$V_{o2} = 50mV$ (open)

$R_s = 1M\Omega$
$R_{in} = ?$

switch is closed

$V_{in1} = V_s$ $\quad 100mV = A_v \cdot V_{in1}$ $\quad \frac{100mV}{50mV} = \frac{V_s}{V_s \cdot \frac{R_{in}}{R_{in}+1M\Omega}}$ $\quad 2$

switch is open

$V_{in2} = V_s \cdot \frac{R_{in}}{R_{in}+1M\Omega}$ $\quad 50mV = A_v \cdot V_{in2}$ $\quad 2 = \frac{R_{in}+1M\Omega}{R_{in}}$

$\boxed{R_{in} = 1M\Omega}$ //

③

$R_L = 100\Omega$
$R_{in} = 200\Omega$

$G_p = 5000$

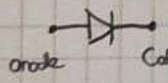
$A_v = A_i \cdot \frac{R_L}{R_{in}}$

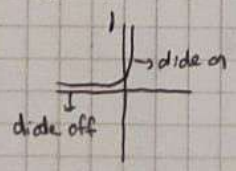
$G_p = A_v \cdot A_i = A_i^2 \cdot \frac{R_L}{R_{in}} \Rightarrow 5000 = (A_i)^2 \frac{100}{200}$ $\quad (A_i)^2 = 10000$ $\quad \boxed{A_i = 100}$ //

$A_v = 100 \cdot \frac{100}{200} = 50$ $\quad \boxed{A_v = 50}$ //

Voltage gain=50
Current gain=100 //

④ Ideal diode → forward bias → 0Ω
→ reverse bias → ∞Ω





diode on → $i_D > 0$

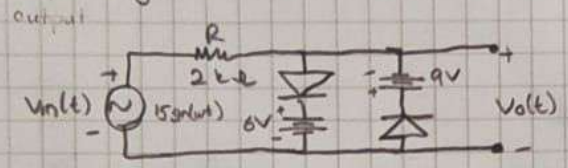
diode off → $V_{anode} < V_{cathode}$ → $V_D < 0$

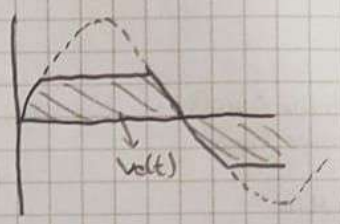
⑤ $A_v = 5$
$R_L = 100\,\Omega$
$R_{in} = 0.01\,M\Omega$

$$A_v = A_i \cdot \frac{R_L}{R_{in}} \rightarrow 5 = A_i \cdot \frac{100}{10000} \quad \boxed{A_i = 500}$$

$$G = A_v \cdot A_i = 500 \cdot 5 = 2500 \quad \boxed{G = 2500}$$

⑥ A clipper circuit clips off part of the input waveform to produce a desired output signal. Diodes can be used to form clipper circuits.

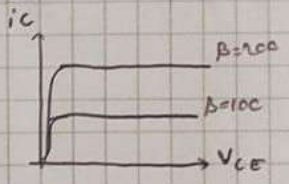




⑦ $\beta = 100$
$i_B = 0.1\,mA$

1/ $i_C = \beta \cdot i_B \rightarrow i_C = 100 \cdot 0.1\,mA = 10\,mA$

2/ $i_C = \beta \cdot i_B \rightarrow i_C = 200 \cdot 0.1\,mA = 20\,mA$



⑧ State of diode

$V_{cathode} = 0V$ $\qquad V_{anode} > V_{cathode} \rightarrow$ on V

$$V_{anode} = \frac{2}{2+1} \cdot (12 - (-16)) - 16 = \left(\frac{2}{3} \cdot 28\right) - 16 = 2.67\,V$$

$$\overset{V_x}{4/} \quad \frac{V_x - 12}{1} + \frac{V_x - (-16)}{2} + \frac{V_x}{4} = 0 \Rightarrow 7V_x = 16 \quad \boxed{V_x = 2.29\,V}$$

$$i_x = \frac{V_x}{4k\Omega} \Rightarrow i_x = \frac{2.29}{4} \Rightarrow \boxed{0.57\,mA}$$

(9) like NPN transistor circuit

$V_{CC} = 12V$

$R_B = 500\,k\Omega$

$R_C = 7.5\,k\Omega$

$V_{BE} = 0.8V$

$V_C = 5V\ (V_{CE})$

$\beta = I_C / I_B$

$V_{CC} = I_B \cdot R_B + V_{BE}$

$12V = I_B \cdot 500\,k\Omega + 0.8V$

$\boxed{I_B = 0.0224mA}$

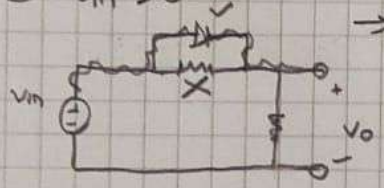
$I_C = \frac{V_{CC} - V_C}{R_C}$

$I_C = \frac{12V - 5V}{7.5k\Omega}$

$\boxed{I_C = 0.933mA}$

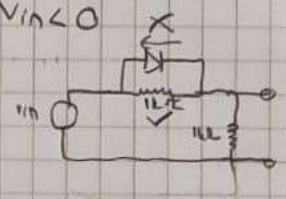
$\beta = \frac{0.933mA}{0.0224mA} = \boxed{41.65}$

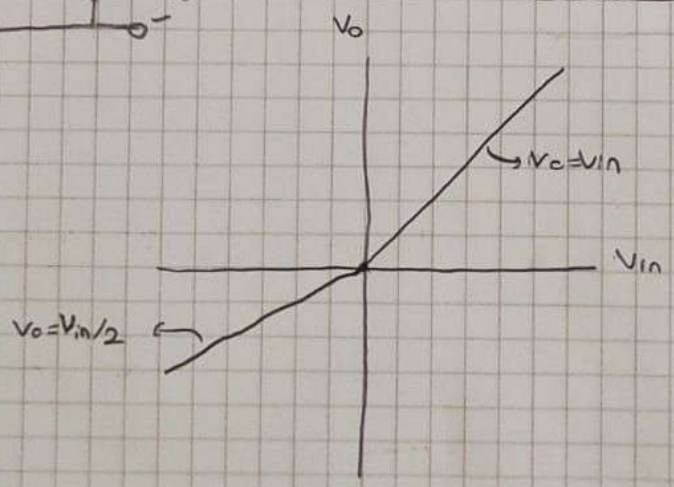
(10) $V_{in} > 0$



$\rightarrow \boxed{V_o = V_{in}}$

$V_{in} < 0$



$V_o = V_{in} \cdot \frac{1k\Omega}{1k\Omega + 1k\Omega}$

$\boxed{V_o = \frac{1}{2} V_{in}}$